



## Q1 Keyword Arguments

5 marks

Write a function `employee(empid, name, salary)` and call the function by changing the order of the arguments using keywords.

### Your Code

```
1 def employee(empid, name, salary):  
2     print("Employee ID:", empid)  
3     print("Name:", name)  
4     print("Salary:", salary)  
5  
6  
7 employee(name="Mani Chandan", empid=192525392, salary=40000)
```

### Input

```
Mani Chandan  
192525392  
400000
```

### Output

```
Employee ID: 192525392  
Name: Mani Chandan  
Salary: 40000
```



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PYTHON PROGRAMMING



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## Q2 Keyword Arguments

5 marks

Write a function book(title, author, price) and display the book details using keyword arguments.

### Your Code

```
1 def book(title, author, price):  
2     print("Book Details")  
3     print("Title:", title)  
4     print("Author:", author)  
5     print("Price:", price)  
6  
7  
8 book(title="Wings if Fire", author="A.P.J. Abdul Kalam", price=700)
```

### Input

Wings of Fire  
A.P.J. Abdul Kalam  
700

### Output

```
Book Details  
Title: Wings if Fire  
Author: A.P.J. Abdul Kalam  
Price: 700
```



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### Q3 Variable-Length Arguments (\*args)

5 marks

Write a function that accepts any number of integers and returns their sum.

#### Your Code

```
1 def add(*args):  
2     return sum(args)  
3  
4  
5 nums = list(map(int, input().split(',')))  
6 print(add(*nums))
```

#### Input

1,2,3,4

#### Output

10



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## Q4 Variable-Length Arguments (\*args)

5 marks

Write a function that accepts any number of marks and returns the average.

### Your Code

```
1 def average_marks(*args):  
2     return sum(args) / len(args)  
3  
4  
5 nums = list(map(int, input().split(',')))  
6 print(average_marks(*nums))
```

### Input

10,20,30,40

### Output

25.0



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## Q5 Variable-Length Arguments (\*args)

5 marks

Write a function that accepts any number of strings and prints each string on a new line.

### Your Code

```
1 def print_strings(*args):  
2     for i in args:  
3         print(i)  
4  
5  
6 s = input().split(' ', '  
7 print_strings(*s)
```

### Input

apple, mango, orange, mango

### Output

```
apple  
mango  
orange  
mango
```



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## Q6 Variable-Length Arguments (\*args)

5 marks

Write a function that accepts any number of numbers and returns the largest value.

### Your Code

```
1 def largest_value(*nums):  
2     return max(nums)  
3  
4 s = input().split(' ')  
5 nums = [int(x) for x in s]  
6 print(largest_value(*nums))
```

### Input

stdin input

### Output

10



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## Q7 Recursive Functions

5 marks

Write a recursive function to generate the Fibonacci series up to n terms.

## Your Code

```
1 def fib(n):
2     if n == 1:
3         return [0]
4     if n == 2:
5         return [0, 1]
6
7     series = fib(n - 1)
8     series.append(series[-1] + series[-2])
9
10    return series
11
12
13 n = int(input().strip())
14 print(*fib(n))
```

## Input

7

## Output

0 1 1 2 3 5 8



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## Q8 Recursive Functions

5 marks

Write a recursive function to find the sum of digits of a number.

## Your Code

```
1 def sum_of_digits(n):  
2     n = abs(n) # Handle negative numbers  
3  
4     if n == 0:  
5         return 0  
6  
7     return (n % 10) + sum_of_digits(n // 10)  
8  
9  
10 # Example usage  
11 num = int(input())  
12 print(sum_of_digits(num))
```

## Input

123

## Output

6



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## Q9 Recursive Functions

5 marks

Write a recursive function to reverse a string.

### Your Code

```
1 def reverse_string(s):  
2     if len(s) == 0:  
3         return s  
4  
5     return reverse_string(s[1:]) + s[0]  
6  
7  
8 text = input()  
9 print(reverse_string(text))
```

### Input

hello

### Output

olleh



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## Q10 Recursive Functions

5 marks

Write a recursive function to calculate x? (power of a number).

## Your Code

```
1 def power(x, n):
2     if n == 0:
3         return 1
4     if n == 1:
5         return x
6
7     if n > 0:
8         return x * power(x, n - 1)
9
10    return 1 / power(x, -n)
11
12
13 x = float(input("Enter x: "))
14 n = int(input("Enter n: "))
15
16 print("Output:", power(x, n))
```

## Input

2  
5

## Output

Enter x: Enter n: Output: 32.0



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